

Wellbore Trajectory and Position-Uncertainty Modelling in Python

Minimum Curvature, Selected ISCWSA Revision 5 Error Propagation,
Analytical Sensitivities, and Three-Dimensional Visualization

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Scope and safety statement. This project is an independent educational and engineering implementation. It evaluates a selected eleven-term subset of the ISCWSA Revision 5 framework using generic benchmark magnitudes. It is not a complete tool-specific Position Uncertainty Model and has not been certified for operational survey acceptance or anti-collision decisions.

Executive Summary

Objective. This project implements a Python workflow from directional-survey stations to well-bore coordinates and three-dimensional position covariance. Position is reconstructed with the Minimum Curvature Method (MCM). Survey uncertainty is propagated for a selected eleven-term ISCWSA Revision 5 subset using both the documented balanced-tangential (BT) sensitivities and analytical derivatives of the complete MCM displacement. Covariance is retained at the 1σ level internally, report figures and numerical axes use an explicitly labelled 2σ convention by default.

Verification evidence. Three complementary checks were used. First, three ISCWSA spreadsheet trajectories were reconstructed with maximum TVD residuals from 4.89 mm to 35.90 mm. Second, BT propagation reproduced ten of the eleven selected covariance terms at approximately floating-point precision, the remaining DSTG element difference was $9.278 \times 10^{-5} \text{ m}^2$ and is consistent with the term's dependence on the slightly different reconstructed TVD. Third, analytical MCM and BT Jacobians agreed with their respective central finite differences to maximum elementwise discrepancies of 1.760×10^{-11} and 1.705×10^{-11} . Against the rounded Volve coordinates for well 15/9-F-11 A, the largest coordinate residual was 8.70 mm.

MCM-versus-BT result. The largest column difference was $1.188 \times 10^{-2} \text{ m/deg}$ for the endpoint inclination sensitivities. After propagation through the common eleven-term model, the maximum difference in the NE 2σ semi-major axis was 1.559 mm, the maximum NE-axis azimuth difference was 0.00419° . The full covariance difference had a global BT-referenced relative RMS of 1.244×10^{-4} . These results show that the sensitivity choice has a small effect for this trajectory and model subset, but they do not establish general MCM–BT equivalence.

Engineering interpretation and scope. Beyond numerical verification, the Volve well is used as a trajectory-based case study to interpret how changes in well attitude and survey spacing correspond to the accumulation, orientation, and local geometry of the modelled uncertainty. The resulting EOW 2σ values are 11.01 m and 3.52 m for the NE semi-axes and $\pm 11.62 \text{ m}$ for the vertical marginal extent. These values illustrate the behaviour of the selected eleven-term model and do not estimate the uncertainty achieved during the actual drilling operation. The visualized cross-hole tube connects local normal-plane covariance rings, it is not a joint three-dimensional confidence volume or an operational anti-collision envelope. The implementation remains an educational framework: it does not contain a complete tool-specific Position Uncertainty Model (PUM), surface or tie-on uncertainty, multi-well correlation, separation-factor logic, or operational certification.

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1 Project Overview

Directional drilling requires both a spatial representation of the well path and an estimate of the uncertainty in the calculated position. The first task maps measured depth, inclination, and azimuth to North, East, and vertical coordinates. The second maps possible survey-measurement errors through the trajectory calculation and accumulates their spatial effects according to their correlation behaviour.

The purpose of this project was to implement and verify these steps in a transparent Python workflow. The work was organised around four engineering questions:

1. Can the implementation reproduce reference well coordinates using the Minimum Curvature Method?
2. Can the covariance engine reproduce the selected ISCWSA benchmark terms when the documented balanced-tangential sensitivities are used?
3. Do independently evaluated central finite differences reproduce the analytical BT and MCM sensitivity matrices?
4. For the selected real-well case, how much does retaining the full MCM ratio-factor sensitivity change the propagated uncertainty relative to BT?

Together, these questions define the numerical-validation and sensitivity-comparison core of the project. The resulting modular pipeline contains survey input checks, trajectory reconstruction, analytical and numerical sensitivity calculations, selected ISCWSA error-term propagation, benchmark verification, reporting, and visualization.

The Volve application then serves two purposes: it provides a coordinate-consistency comparison and illustrates how changes in inclination, azimuth, dogleg severity, and survey spacing relate to the calculated covariance response. This second use is interpretive rather than an additional numerical benchmark or controlled causal study.

The repository is intended to demonstrate numerical reasoning, verification practice, engineering interpretation, and familiarity with wellbore-positioning concepts. It does not replace a service-provider tool code or operational survey program.

1.1 Main demonstrated results

- Three ISCWSA trajectory examples were reconstructed with maximum TVD residuals between 4.9 mm and 35.9 mm.
- The Volve 15/9-F-11 A coordinates were reproduced with a maximum component residual of 8.70 mm against the rounded source data.
- BT propagation reproduced ten out of eleven selected benchmark terms to approximately machine precision.
- Analytical BT and MCM sensitivities agreed with corresponding central finite-difference calculations along the tested well path.
- The maximum difference in the projected semi-major uncertainty axis between MCM and BT propagation was 1.55 mm, or approximately 0.015%, for the selected well and error subset.
- The Volve trajectory was segmented using changes in inclination, azimuth, DLS and survey spacing to support an engineering interpretation of the calculated covariance response. The observed associations are treated as descriptive rather than as isolated causal effects.

2 Data, Coordinates, and Scope

2.1 Survey representation

A survey station contains measured depth D , inclination I , and azimuth A . Inclination is measured from the downward vertical. Azimuth is measured clockwise from the North reference used by the input survey. Internally, angles are converted to radians before trigonometric evaluation.

The Earth-centred coordinate order is North–East–Vertical (NEV), where the vertical coordinate is positive downward and is reported as True Vertical Depth (TVD). The unit tangent vector at a survey station is

$$\mathbf{t}(I, A) = \begin{bmatrix} \sin I \cos A \\ \sin I \sin A \\ \cos I \end{bmatrix}. \quad (1)$$

The implementation assumes that input azimuths already use the same north reference as the requested NEV coordinates. Magnetic-to-true or true-to-grid corrections, convergence uncertainty, and datum transformation are outside the present scope.

2.2 Datasets

The study uses two data sources:

- Three ISCWSA Revision 5 spreadsheet examples, which provide survey profiles and station-by-station covariance diagnostics for individual error sources [1, 2].
- The public Equinor Volve dataset for well 15/9-F-11 A, which contains measured depth, inclination, azimuth, TVD, North, and East coordinates [5].

The two datasets have different roles. The ISCWSA spreadsheets provide numerical implementation references for the selected trajectory and covariance calculations. The Volve data are used both for a coordinate-consistency comparison and as a real-well case study for interpreting the trajectory-dependent development of the modelled uncertainty.

Neither dataset provides independent physical ground truth. The Volve coordinates are rounded, and the uncertainty case uses generic benchmark error magnitudes rather than a tool-specific model for the survey system used during the actual drilling operation.

2.3 Input checks

The survey preprocessing stage verifies that required columns are present, values are numeric and finite, measured depth is increasing, inclination lies within 0° to 180° , and azimuth is normalized to a consistent 360° interval. Repeated rows with identical measured depth, inclination, and azimuth are collapsed with a message. A repeated measured depth with a conflicting attitude, or any decrease in measured depth, is rejected.

The current project also flags very short survey intervals. A short interval is not inherently invalid, so any rejection threshold is treated as a configurable data-quality policy rather than an ISCWSA requirement. Operational survey data would require a project-specific QC policy and retention of all valid survey stations.

3 Trajectory Calculation

3.1 Dogleg angle

For two consecutive station tangents, the dogleg angle β is the angle between $\mathbf{t}_1$ and $\mathbf{t}_2$. Direct evaluation with $\arccos(\mathbf{t}_1 \cdot \mathbf{t}_2)$ loses resolution when the vectors are nearly parallel. The implementation therefore uses the equivalent Haversine form

$$\beta = 2 \arcsin \left[\sqrt{\sin^2 \left(\frac{I_2 - I_1}{2} \right) + \sin I_1 \sin I_2 \sin^2 \left(\frac{A_2 - A_1}{2} \right)} \right]. \quad (2)$$

The argument of the square root is clipped within floating-point tolerance to the interval $[0, 1]$ to guard numerical round off. Equation (2) also shows why azimuth changes have little geometric effect close to vertical: their contribution is multiplied by $\sin I_1 \sin I_2$.

3.2 Dogleg severity

When β is evaluated in radians, dogleg severity in degrees per reference length L_{ref} is

$$\text{DLS} = \beta \frac{180^\circ}{\pi} \frac{L_{\text{ref}}}{\Delta D}. \quad (3)$$

In the metric outputs of this project, $L_{\text{ref}} = 30 \text{ m}$.

3.3 Minimum Curvature Method

The Minimum Curvature Method assumes that the path between two survey stations is a circular arc in the plane defined by the endpoint tangents [4]. For course length $L = \Delta D$, the ratio factor is

$$R(\beta) = \frac{2}{\beta} \tan \left(\frac{\beta}{2} \right), \quad (4)$$

and the displacement is

$$\Delta \mathbf{r}_{\text{MCM}} = \frac{L}{2} (\mathbf{t}_1 + \mathbf{t}_2) R(\beta). \quad (5)$$

For small doglegs, direct evaluation of Equation (4) is replaced by

$$R(\beta) = 1 + \frac{\beta^2}{12} + \frac{\beta^4}{120} + O(\beta^6). \quad (6)$$

The Cartesian coordinates are obtained by cumulatively summing the displacement vectors from the initial position.

3.4 Balanced Tangential displacement

The balanced-tangential displacement is the mean of the endpoint tangents multiplied by course length:

$$\Delta \mathbf{r}_{\text{BT}} = \frac{L}{2} (\mathbf{t}_1 + \mathbf{t}_2). \quad (7)$$

The ISCWSA mathematical definition recommends MCM for calculating position but uses the simpler BT derivatives for the position sensitivity in the uncertainty calculation [1]. This project retains that documented BT path and implements an additional analytical MCM sensitivity for comparison.

4 Analytical Position Sensitivities

4.1 Segment Jacobian

For one survey interval, define

$$\mathbf{p} = [I_1 \quad A_1 \quad I_2 \quad A_2 \quad L]^\top. \quad (8)$$

The segment Jacobian is

$$\mathbf{J}_{\text{seg}} = \frac{\partial \Delta \mathbf{r}}{\partial \mathbf{p}}, \quad \mathbf{J}_{\text{seg}} \in \mathbb{R}^{3 \times 5}. \quad (9)$$

The derivatives of the tangent vector are

$$\frac{\partial \mathbf{t}}{\partial I} = \begin{bmatrix} \cos I \cos A \\ \cos I \sin A \\ -\sin I \end{bmatrix}, \quad \frac{\partial \mathbf{t}}{\partial A} = \begin{bmatrix} -\sin I \sin A \\ \sin I \cos A \\ 0 \end{bmatrix}. \quad (10)$$

Let

$$\mathbf{s} = \mathbf{t}_1 + \mathbf{t}_2, \quad q = \mathbf{t}_1^\top \mathbf{t}_2 = \cos \beta. \quad (11)$$

For any endpoint angle $x \in \{I_1, A_1, I_2, A_2\}$, the analytical MCM derivative can be written compactly as

$$\frac{\partial \Delta \mathbf{r}_{\text{MCM}}}{\partial x} = \frac{L}{2} \left[R \frac{\partial \mathbf{s}}{\partial x} + \mathbf{s} \frac{\partial R}{\partial q} \frac{\partial q}{\partial x} \right]. \quad (12)$$

Here, $\partial q / \partial x$ is obtained by differentiating the corresponding endpoint tangent. For example,

$$\frac{\partial q}{\partial I_1} = \left(\frac{\partial \mathbf{t}_1}{\partial I_1} \right)^\top \mathbf{t}_2. \quad (13)$$

For finite dogleg angle,

$$\frac{\partial R}{\partial q} = \frac{2 \tan(\beta/2) - \beta \sec^2(\beta/2)}{\beta^2 \sin \beta}. \quad (14)$$

Near zero dogleg, the stable limit used in the implementation is

$$\frac{\partial R}{\partial q} = -\frac{1}{6} - \frac{11}{180} \beta^2 + O(\beta^4). \quad (15)$$

The course-length derivative is

$$\frac{\partial \Delta \mathbf{r}_{\text{MCM}}}{\partial L} = \frac{1}{2} \mathbf{s} R. \quad (16)$$

The BT Jacobian follows by setting $R = 1$ and $\partial R / \partial q = 0$ in Equations (12) and (16).

4.2 Connection to station sensitivities

The 3×5 matrix in Equation (9) describes one interval. The ISCWSA calculation is station based: a survey measurement at an interior station affects both the interval that precedes it and the interval that follows it. Define

$$\mathbf{G}_k^- = \frac{\partial \Delta \mathbf{r}_k}{\partial (D_k, I_k, A_k)}, \quad \mathbf{G}_k^+ = \frac{\partial \Delta \mathbf{r}_{k+1}}{\partial (D_k, I_k, A_k)}. \quad (17)$$

The interior station sensitivity is then

$$\mathbf{G}_k = \mathbf{G}_k^- + \mathbf{G}_k^+. \quad (18)$$

Measured depth requires particular care: increasing D_k increases the preceding course length and decreases the following course length. At the final station of interest, only the preceding-interval contribution remains. This station-to-segment mapping is why the implementation stores both endpoint contributions rather than classifying errors only by measurement axis.

5 Selected ISCWSA Error Propagation

5.1 Error-vector construction

For error source i at station k , let σ_i be the one-standard-deviation magnitude and let

$$\mathbf{w}_{i,k} = \frac{\partial(D, I, A)}{\partial \varepsilon_i} \quad (19)$$

be its measurement-space weighting vector. The corresponding NEV position error vector is

$$\mathbf{e}_{i,k} = \sigma_i \mathbf{G}_k \mathbf{w}_{i,k}. \quad (20)$$

At the final station of interest, the endpoint vector is evaluated using the preceding interval only:

$$\mathbf{e}_{i,K}^* = \sigma_i \mathbf{G}_K^- \mathbf{w}_{i,K}. \quad (21)$$

Equations (20) and (21) are evaluated with either the documented BT sensitivities or the analytical MCM sensitivities. The weighting functions and propagation mode are otherwise unchanged.

5.2 Propagation modes

The ISCWSA framework distinguishes random (R), systematic (S), well-by-well (W), and global (G) correlation behaviour.

R – Random. Contributions are treated as independent between survey stations, survey legs, and wells.

S – Systematic. Contributions are correlated between stations within the same survey leg but are independent between different legs and wells.

W – Well-by-well. Contributions are correlated between stations and survey legs within the same well but are independent between different wells.

G – Global. Contributions are correlated between stations, survey legs, and wells within the applicable field or common reference context.

For the single-leg, single-well calculation implemented here, the S, W, and G modes have the same numerical accumulation form. Their distinctions become important when multiple survey legs or wells are evaluated.

Random source. Independent station contributions are accumulated as outer products,

$$\mathbf{C}_{i,K}^R = \sum_{k=1}^{K-1} \mathbf{e}_{i,k} \mathbf{e}_{i,k}^\top + \mathbf{e}_{i,K}^* (\mathbf{e}_{i,K}^*)^\top. \quad (22)$$

Correlated source within the evaluated leg. The error vectors are summed before the outer product,

$$\mathbf{C}_{i,K}^{\text{corr}} = \mathbf{E}_{i,K} \mathbf{E}_{i,K}^\top, \quad \mathbf{E}_{i,K} = \sum_{k=1}^{K-1} \mathbf{e}_{i,k} + \mathbf{e}_{i,K}^*. \quad (23)$$

Systematic, well-by-well, and global sources have different correlation semantics across survey legs and wells even where their single-leg numerical form is identical. Multi-leg tie-on handling, well-by-well propagation, and inter-well global correlations are not implemented in this version.

Assuming independence between distinct error sources, the total covariance is

$$\mathbf{C}_K = \sum_{i=1}^M \mathbf{C}_{i,K}. \quad (24)$$

The NEV covariance has the form

$$\mathbf{C} = \begin{bmatrix} C_{NN} & C_{NE} & C_{NV} \\ C_{NE} & C_{EE} & C_{EV} \\ C_{NV} & C_{EV} & C_{VV} \end{bmatrix}. \quad (25)$$

The diagonal entries are coordinate variances. The off-diagonal entries describe linear covariance and determine the orientation of the principal uncertainty axes.

5.3 Incremental accumulation

A direct implementation that re-sums all previous station contributions at every station has quadratic cost in the number of stations. The project instead stores the running vector sums and covariance sums required by Equations (22) and (23). Each new station therefore requires constant work per error source. For n stations and M error sources, covariance accumulation scales as $O(nM)$ and becomes $O(n)$ when M is fixed.

This recurrence is an implementation characteristic rather than a claimed new ISCWSA method.

5.4 Implemented benchmark subset

Table 1 lists the eleven terms used in the current validation configuration. The magnitudes are the values extracted for the selected benchmark setup, they do not describe the actual tool used in the Volve well. The official Revision 5 framework contains additional possible sources, and an operational Position Uncertainty Model selects appropriate terms, magnitudes, and correlation modes for the actual survey system.

Table 1: Selected benchmark error terms. D denotes measured depth and V denotes TVD. DEC-U is classified as well-by-well in the Revision 5 definition, within this single-well calculation its correlated accumulation has the same numerical form as a global term.

Term	Input	Mode	Magnitude	Unit	Weight
DRFR	MD	R	0.35	m	1
DSFS	MD	S	5.6×10^{-4}	–	D
DSTG	MD	G	2.5×10^{-7}	m^{-1}	$D V$
DEC-U	Az	W	0.16	deg	1
DEC-OS	Az	G	0.24	deg	1
DEC-OH	Az	G	0.21	deg	1
DEC-OI	Az	G	0.05	deg	1
DECR	Az	R	0.10	deg	1
SAGE	Inc	S	0.20	deg	$(\sin I)^{0.25}$
XYM1	Inc	S	0.10	deg	$ \sin I $
XYM2	Az	S	0.10	deg	–1

The term names follow the benchmark workbook naming used by the implementation. For depth terms, Table 1 lists the cumulative weighting function $W(D,V)$. The segment implementation applies $\Delta W_k = W_k - W_{k-1}$: ΔD for DSFS and $\Delta(DV)$ for DSTG. DRFR is evaluated using the corresponding station-depth differencing logic.

Table 2: Short descriptions of the eleven selected error sources. The descriptions identify the uncertainty represented by each term; magnitudes and propagation modes are taken from the selected PUM.

Term	Description
DRFR	Random uncertainty in the measured-depth reference.
DSFS	Systematic measured-depth scale-factor error whose effect increases with measured depth.
DSTG	Global drill-pipe depth-stretch error weighted by measured depth and TVD.
DEC-U	Well-specific declination uncertainty treated as uncorrelated between different wells.
DEC-OS	Declination uncertainty associated with crustal magnetic-field content omitted by standard-definition geomagnetic models.
DEC-OH	Declination uncertainty associated with crustal magnetic-field content omitted by high-definition geomagnetic models.
DEC-OI	Declination uncertainty associated with crustal magnetic-field content omitted by an in-field referencing model.
DECR	Random declination uncertainty representing short-term or station-dependent variation in the magnetic reference.
SAGE	Enhanced sag uncertainty representing gravity-induced deflection between the survey tool or BHA and the borehole direction.
XYM1	First systematic transverse borehole-misalignment component, acting through the inclination weighting $ \sin I $.
XYM2	Second systematic transverse borehole-misalignment component, acting through the azimuth weighting –1.

6 Uncertainty Scaling and Visualization

6.1 Primary 2σ convention

For a selected plane, the corresponding 2×2 marginal covariance matrix is extracted from Equation (25). Because it is symmetric, its eigenvectors define the ellipse directions and its eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$ define the squared one-standard-deviation axis lengths.

The implementation stores covariance at the 1σ level and uses a 2σ display convention by default. With a selected sigma multiplier s , the semi-axes are

$$a = s\sqrt{\lambda_1}, \quad b = s\sqrt{\lambda_2}, \quad s = 2 \text{ by default.} \quad (26)$$

This is an axis-scaling convention, not a claim of 95% joint coverage in every dimension. For a bivariate Gaussian, the region bounded by the 2σ ellipse has approximately 86.47% joint coverage.

6.2 Optional joint-probability convention

When a joint probability p is required instead, the code can use

$$k_d(p) = \sqrt{\chi_d^2(p)}, \quad a_j = k_d(p)\sqrt{\lambda_j}, \quad (27)$$

where d is the dimension of the region. At $p = 0.95$, the factors are $k_1 = 1.9600$, $k_2 = 2.4477$, and $k_3 = 2.7955$. Consequently, a 95% univariate interval, a 95% joint planar ellipse, and a 95% joint spatial ellipsoid must not be given the same multiplier. The NE, NV, and EV plots are marginal two-dimensional projections and use $d = 2$ when probability mode is selected.

6.3 Three-dimensional representation

A full three-dimensional Gaussian confidence ellipsoid would use k_3 in probability mode. The displayed uncertainty tube is instead constructed from stationwise two-dimensional covariance marginals in the plane normal to the local well tangent. Its rings use $s = 2$ by default, or k_2 in probability mode. The tube is therefore a geometric visualization of local cross-hole uncertainty rings, not a single three-dimensional probability region.

The NEV covariance is rotated into a local High-side–Lateral–Along-hole frame and its 2×2 normal-plane marginal is decomposed. The resulting rings are rotated back to NEV and lofted along the trajectory.

6.4 Interpolation

Tangent directions between physical survey stations are interpolated using spherical linear interpolation (SLERP), consistent with the circular-arc assumption of MCM. For a fraction $f \in [0, 1]$,

$$\mathbf{t}(f) = \frac{\sin[(1-f)\beta]}{\sin\beta} \mathbf{t}_1 + \frac{\sin(f\beta)}{\sin\beta} \mathbf{t}_2. \quad (28)$$

Near zero dogleg, normalized linear interpolation is used. Covariances between physical stations are linearly interpolated only to draw continuous surfaces. Synthetic points are tagged and excluded from stationwise validation and uncertainty accumulation.

7 Verification Method

7.1 Trajectory comparison

For every available reference coordinate y_k^{ref} , the residual is

$$r_k = y_k^{\text{calc}} - y_k^{\text{ref}}. \quad (29)$$

Reported metrics include the mean residual, mean absolute error, root-mean-square residual, maximum absolute residual, EOW residual, and the slope of a descriptive least-squares fit against measured depth. These metrics quantify agreement with the supplied coordinates. They do not establish physical survey accuracy.

7.2 Covariance benchmark

For each selected error term, every unique element of the symmetric NEV covariance matrix is compared at every benchmark station. Both MCM and BT sensitivities are propagated through the same covariance engine. This isolates the effect of the position-sensitivity formulation from the remaining accumulation logic.

7.3 Central finite differences

For parameter p_j , the central finite-difference column is

$$\mathbf{J}_{:,j}^{\text{FD}} = \frac{\Delta \mathbf{r}(\mathbf{p} + h_j \mathbf{u}_j) - \Delta \mathbf{r}(\mathbf{p} - h_j \mathbf{u}_j)}{2h_j}, \quad (30)$$

where $\mathbf{u}_j$ selects the perturbed parameter. The selected perturbations are

$$h_I = h_A = 10^{-3} \text{ deg}, \quad h_D = 0.1 \text{ m}. \quad (31)$$

A convergence sweep is performed at representative intervals to verify the expected balance between round-off error at small h and truncation error at large h . Finite differences verify the implementation of the analytical derivatives, because they use the same displacement function, they do not by themselves validate every modelling assumption.

8 Results

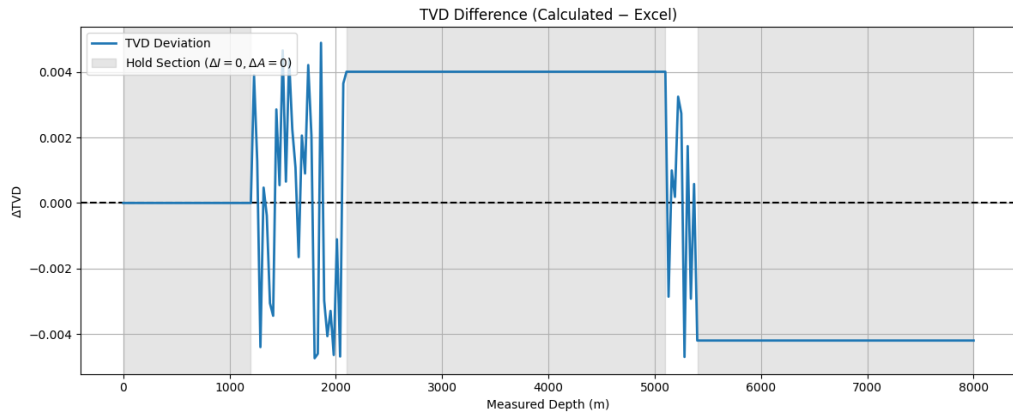
8.1 ISCWSA trajectory examples

Table 3 reports the maximum and EOW TVD residuals for the three spreadsheet examples. The maximum observed discrepancy is 35.9 mm over a 4030 m survey.

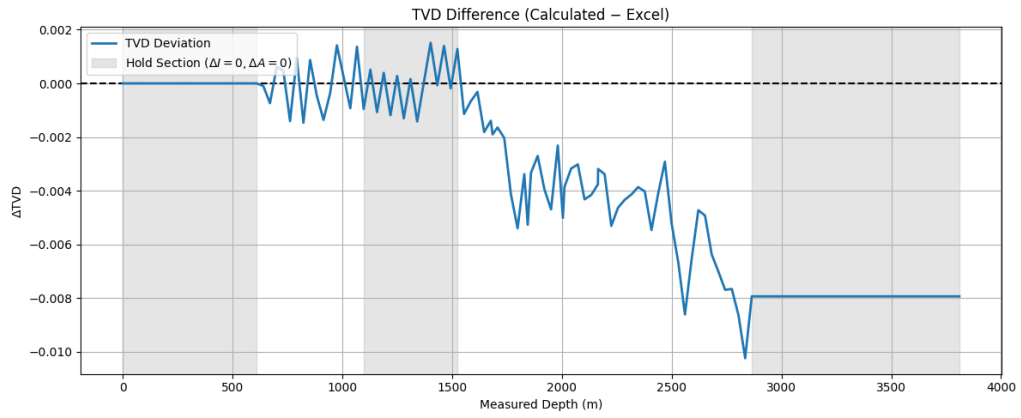
Table 3: Calculated-minus-reference TVD residuals for the three ISCWSA spreadsheet examples at maximum absolute deviation and EOW.

Example	Max. abs. (m)	MD at max. (m)	EOW (m)	Total MD (m)
ISCWSA-1	0.00489	1860.00	−0.004202	8000.00
ISCWSA-2	0.01025	2834.64	−0.007938	3810.00
ISCWSA-3	0.03590	2430.00	+0.02834	4030.00

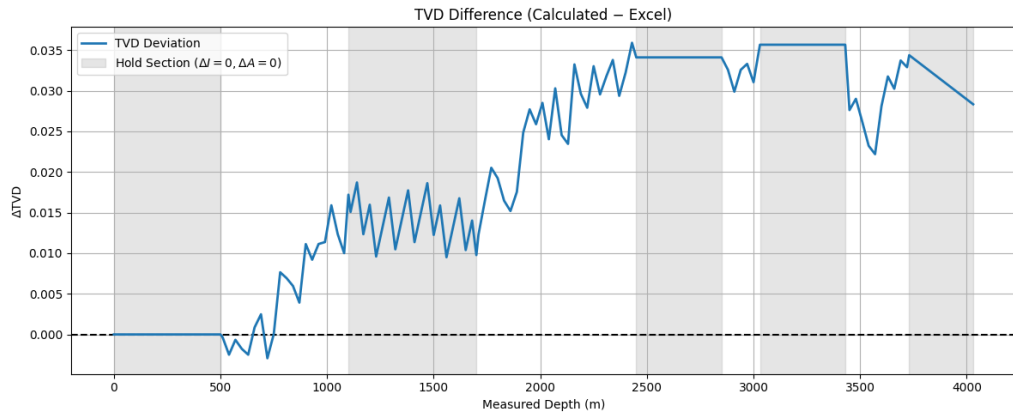
Changing from the Haversine dogleg to the spherical cosine expression and reducing the small-dogleg Taylor threshold did not materially change these residuals. BT trajectory integration and an inclination-only dogleg calculation produced larger deviations. Because the spreadsheet does not expose its trajectory calculation, the remaining TVD residuals cannot be attributed uniquely.



(a) ISCWSA example 1



(b) ISCWSA example 2



(c) ISCWSA example 3

Figure 1: TVD residuals for the three ISCWSA examples.

8.2 Covariance benchmark

Table 4 condenses the stationwise validation by reporting the largest absolute element difference over all three test wells for each term. The MCM column does not represent a failure of the covariance engine: it results from intentionally substituting a different position sensitivity from the BT sensitivity used in the reference framework. When BT sensitivities are used, ten terms agree at approximately floating-point precision.

Table 4: Maximum absolute individual covariance-element difference over the three ISCWSA examples.

Term	Analytical MCM (m ²)	Analytical BT (m ²)
DEC-OS	2.990×10^{-2}	3.553×10^{-15}
DEC-OH	2.289×10^{-2}	8.882×10^{-16}
DEC-U	1.329×10^{-2}	8.882×10^{-16}
SAGE	2.723×10^{-2}	1.421×10^{-14}
XYM2	5.191×10^{-3}	7.105×10^{-15}
XYM1	5.106×10^{-3}	4.405×10^{-13}
DSTG	3.013×10^{-3}	9.278×10^{-5}
DEC-OI	1.298×10^{-3}	1.776×10^{-15}
DSFS	1.299×10^{-3}	3.375×10^{-14}
DRFR	1.543×10^{-3}	4.163×10^{-17}
DECR	9.986×10^{-5}	2.082×10^{-17}

Excluding DSTG, the largest BT global relative RMS across the selected terms was 7.494×10^{-12} . DSTG uses the depth-dependent weighting function $D \cdot TVD$. The small residual, with an maximum relative RMS difference of 0.0132%, in that term is consistent with propagating the reconstructed TVD rather than the slightly different spreadsheet TVD. This explanation is plausible and quantitatively consistent with the result, but the spreadsheet alone does not prove a unique cause.

The BT comparison verifies that the selected weight functions, station contributions, propagation modes, endpoint handling, and covariance assembly can reproduce the reference values for the tested configuration.

8.3 Volve coordinate comparison

Table 5 summarizes agreement between the MCM trajectory and the provided coordinates for well 15/9-F-11 A.

Table 5: Calculated-minus-reference coordinate residuals for Volve well 15/9-F-11 A.

Axis	Mean (m)	MAE (m)	RMSE (m)	Max. abs. (m)	EOW (m)
TVD	+0.00013	0.00232	0.00274	0.00599	+0.00155
N	−0.00052	0.00249	0.00286	0.00576	−0.00576
E	−0.00369	0.00385	0.00468	0.00870	−0.00514

The North and TVD mean residuals are below 0.6 mm in magnitude. The East residual has a mean offset of −3.69 mm. The source coordinates are rounded to 0.01 m, so the high-frequency residual pattern is consistent with reference quantization. The comparison establishes numerical consistency with the supplied data, it is not an operational survey acceptance test.

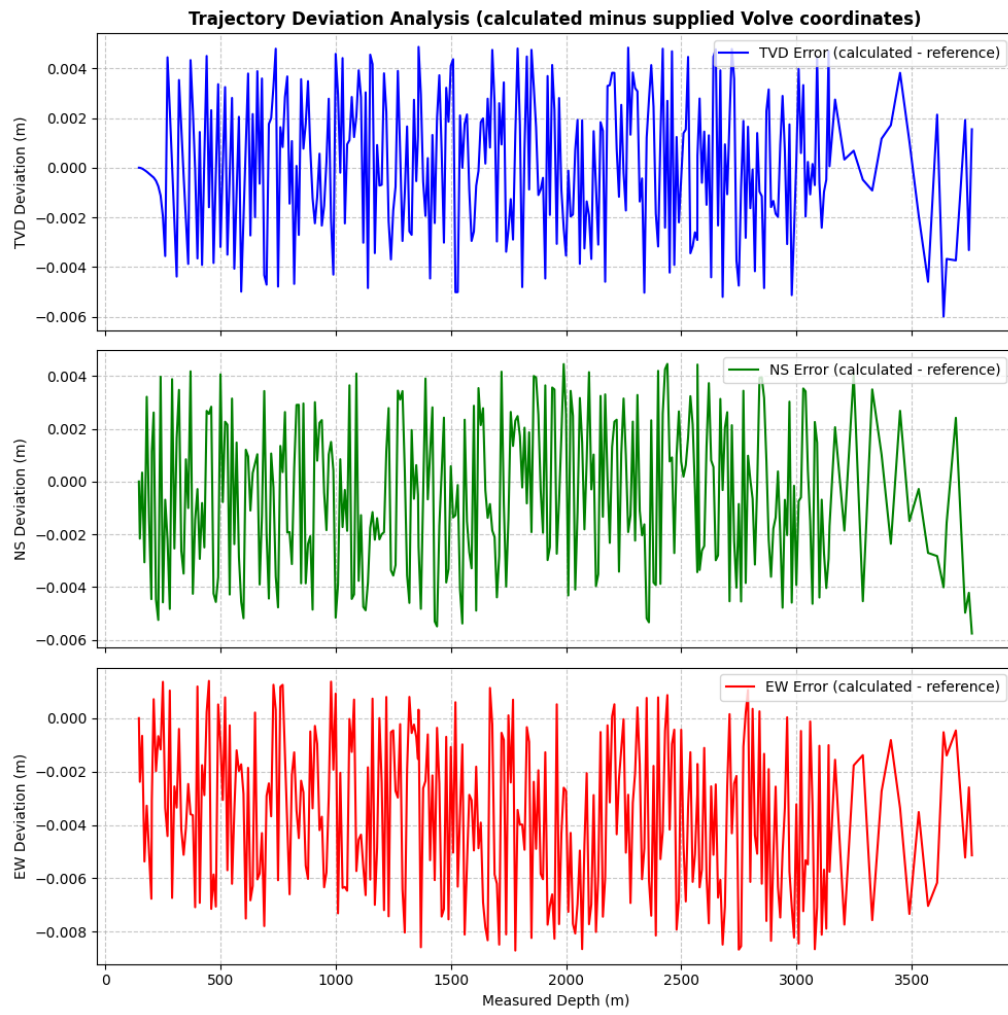
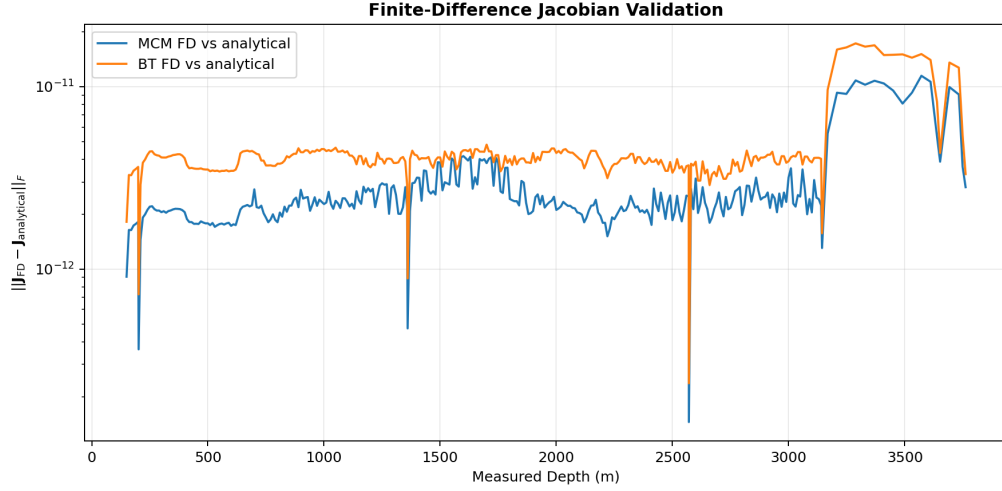


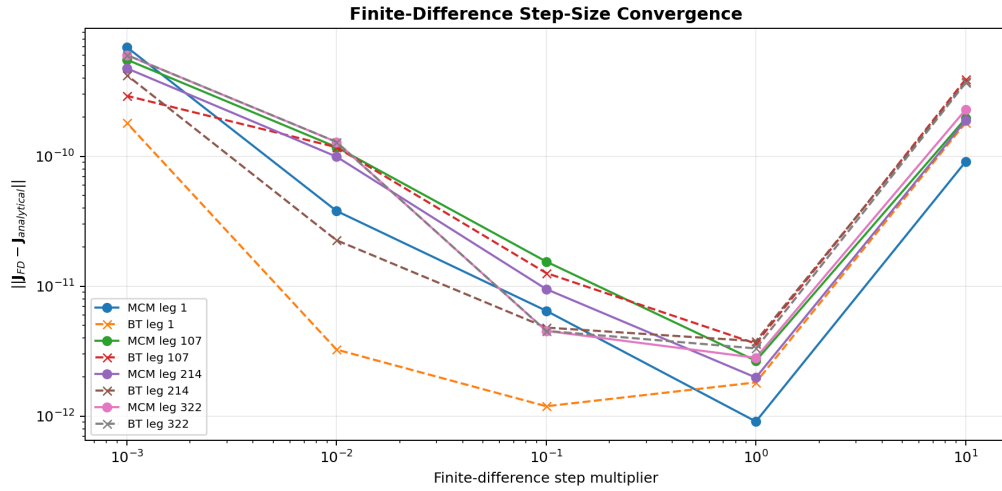
Figure 2: Coordinate residuals versus measured depth for Volve well 15/9-F-11 A.

8.4 Finite-difference verification

Figure 3a compares each analytical sensitivity formulation with the corresponding central finite-difference calculation. The observed agreement is at the numerical scale expected from the selected perturbations. The step-size sweep in Figure 3b shows the expected minimum between round-off and truncation error and supports the perturbations listed below Equation (30).



(a) Analytical versus finite-difference sensitivities.



(b) Perturbation step-size convergence.

Figure 3: Finite-difference verification of the BT and MCM analytical sensitivities.

8.5 MCM versus BT segment sensitivities

The direct comparison is performed column by column because the four angular columns have units of metres per degree, whereas the course-length column has units of metres per metre. For interval k and parameter column j , the absolute sample is

$$d_{k,j} = \|\mathbf{J}_{k,j}^{\text{MCM}} - \mathbf{J}_{k,j}^{\text{BT}}\|_2. \quad (32)$$

Table 6 reports the RMS, 95th percentile, and maximum of $d_{k,j}$ over all physical survey intervals. The relative RMS is a single BT-referenced root-sum-square ratio for each column. This construction retains the native units and avoids cancellation among the NEV components.

Table 6: Column-wise comparison of analytical Minimum Curvature and Balanced Tangential segment sensitivities. Each absolute statistic is calculated from the Euclidean norm of the NEV difference vector for one parameter column.

Column	Unit	RMS	95th percentile	Maximum	Relative RMS	MD at max. (m)
I_1	m/deg	1.402×10^{-3}	7.635×10^{-4}	1.188×10^{-2}	1.253×10^{-2}	3329.40
A_1	m/deg	2.463×10^{-4}	3.313×10^{-4}	2.834×10^{-3}	3.536×10^{-3}	3410.20
I_2	m/deg	1.400×10^{-3}	7.641×10^{-4}	1.188×10^{-2}	1.252×10^{-2}	3329.40
A_2	m/deg	2.462×10^{-4}	3.232×10^{-4}	2.834×10^{-3}	3.601×10^{-3}	3410.20
ΔMD	m/m	7.982×10^{-5}	6.105×10^{-5}	8.394×10^{-4}	7.982×10^{-5}	3329.40

The start- and end-inclination columns behave nearly symmetrically. Both reach a maximum difference of 1.188×10^{-2} m/deg at 3329.40 m MD and have relative RMS values of approximately 1.25%. The azimuth-column differences are smaller, with maxima of 2.834×10^{-3} m/deg. The course-length column has a relative RMS of 7.982×10^{-5} . For the inclination columns, the maximum is much larger than the 95th percentile, showing that the largest differences are localized rather than representative of the complete well.

The maxima occur after the survey spacing changes from approximately 10 m to approximately 40 m. This is consistent with the ratio-factor correction increasing with dogleg, but the field trajectory does not isolate survey interval, dogleg, inclination, and azimuth change as independent causes. A controlled synthetic study would be required for that causal separation.

8.6 Effect on propagated uncertainty

The practical comparison propagates both sensitivity formulations through the same eleven-term model. Table 7 extends the earlier semi-major-axis check to the NE ellipse, vertical marginal extent, all three principal 3D axes, and the complete covariance matrix. Absolute and EOW differences are MCM minus BT except where a magnitude is explicitly reported. The global relative RMS uses the complete BT profile as one reference scale, it is therefore not destabilized by individual values close to zero.

Ellipse orientations are axial rather than directional, so angles separated by 180° represent the same major axis. The reported difference is therefore

$$\Delta\theta_k = |[\theta_k^{\text{MCM}} - \theta_k^{\text{BT}} + 90^\circ] \bmod 180^\circ - 90^\circ|.$$

Orientation statistics are evaluated only where both NE ellipses satisfy

$$\frac{a-b}{a} \geq 0.01,$$

because the major-axis direction becomes ill-conditioned as the ellipse approaches circularity. A relative RMS is not reported for orientation because an axial angle has no physically meaningful multiplicative reference scale.

Table 7: Effect of MCM rather than BT sensitivities on derived 2σ uncertainty outputs. Relative RMS is the RMS MCM–BT difference over all stations divided by the RMS BT reference profile over the same stations. Orientation differences are axial and are reported only where both NE ellipses have relative anisotropy of at least 0.01.

Output	Unit	RMS	Maximum	Relative RMS	EOW difference	MD at max. (m)
NE semi-major axis	m	3.020×10^{-4}	1.559×10^{-3}	5.493×10^{-5}	1.559×10^{-3}	3762.00
NE semi-minor axis	m	1.669×10^{-4}	9.187×10^{-4}	4.828×10^{-5}	9.187×10^{-4}	3762.00
NE major-axis azimuth	°	8.300×10^{-4}	4.187×10^{-3}	—	2.842×10^{-3}	3531.10
Vertical marginal extent	m	2.111×10^{-4}	3.897×10^{-4}	4.754×10^{-5}	2.253×10^{-4}	3168.10
Smallest 3D principal semi-axis	m	1.548×10^{-4}	7.540×10^{-4}	6.828×10^{-5}	7.540×10^{-4}	3762.00
Middle 3D principal semi-axis	m	2.536×10^{-4}	1.272×10^{-3}	6.425×10^{-5}	1.252×10^{-3}	3571.70
Largest 3D principal semi-axis	m	1.798×10^{-4}	6.986×10^{-4}	2.802×10^{-5}	6.986×10^{-4}	3762.00
Full covariance matrix	m ²	1.872×10^{-3}	9.533×10^{-3}	1.244×10^{-4}	9.533×10^{-3}	3762.00

The largest change in any reported 2σ axis is the 1.559 mm NE semi-major-axis difference at EOW. The corresponding global relative RMS is 5.493×10^{-5} , while the earlier maximum pointwise relative difference is approximately 0.015%; these are different summary statistics and should not be interchanged. The NE semi-minor, vertical, and three principal 3D axes all remain within 1.272 mm except for the semi-major value. The maximum axial orientation difference is 0.004187° . Orientation is reported only where both NE ellipses have at least 1% relative anisotropy, because the major-axis direction becomes ill-conditioned as an ellipse approaches circularity.

At matrix level, the maximum stationwise Frobenius difference is $9.533 \times 10^{-3} \text{ m}^2$ at EOW, and the global BT-referenced relative RMS is 1.244×10^{-4} (0.01244%). The six independent covariance elements are reported separately in Table 10. Together, the axis, orientation, and covariance comparisons support a small sensitivity-formulation effect for this well and selected error subset. They do not establish equivalence for other trajectories or PUMs.

Figure 4 retains the stationwise structure behind the summary table. Its panels are intentionally separated by physical unit: the first compares absolute 2σ axis-length differences in metres, the second shows the full covariance Frobenius difference in square metres, and the third shows the wrapped axial NE-angle difference in degrees. The three quantities must not be combined into one scalar or compared by vertical scale.

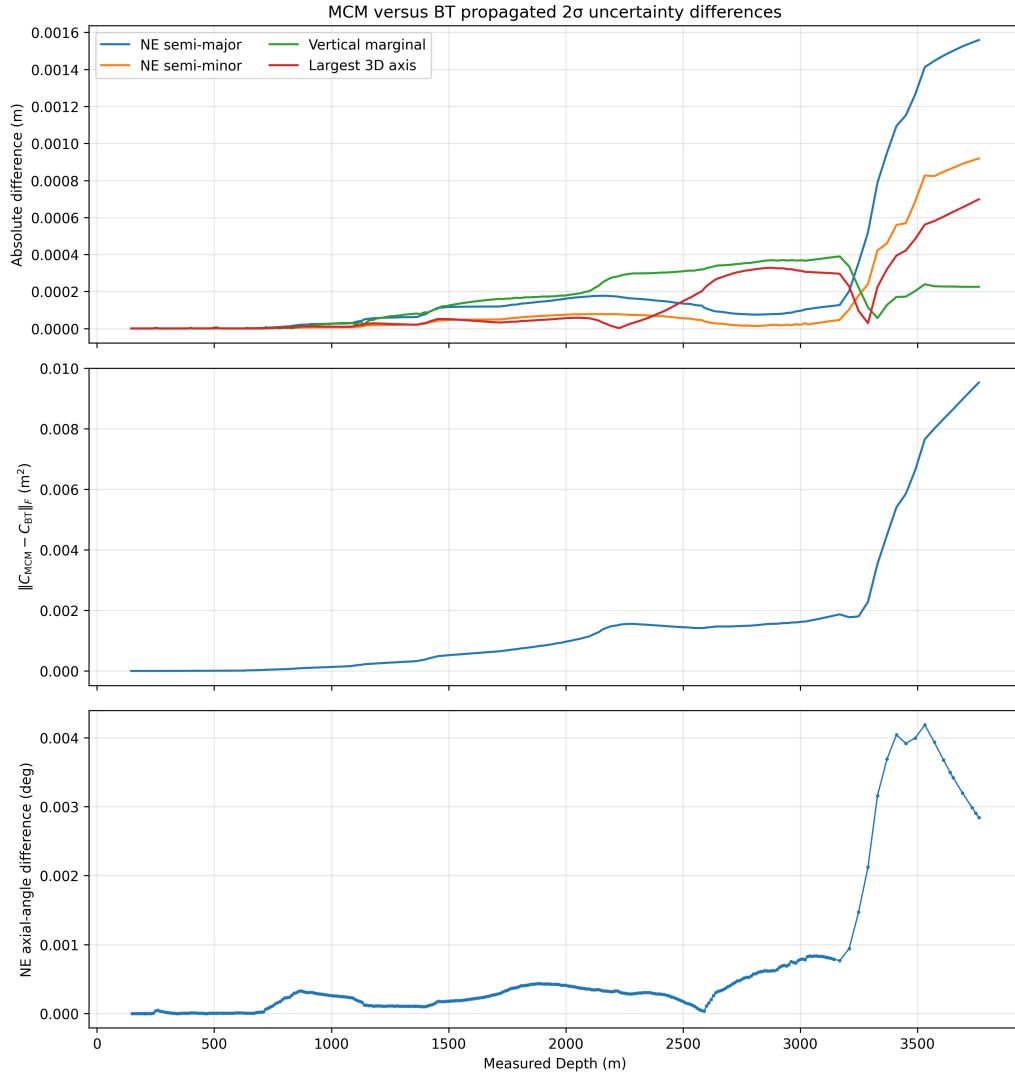


Figure 4: Stationwise effect of substituting analytical MCM sensitivities for BT sensitivities. Axis-length differences, the covariance-matrix Frobenius difference, and the axial NE-orientation difference are shown in separate panels because they have different units and interpretations.

9 Discussion: Physical Interpretation Along the Volve Trajectory

The preceding sections establish the numerical behaviour of the implementation and answer the four engineering questions defined in Section 1. This section uses the resulting Volve calculation to interpret how the modelled uncertainty develops along a realistic directional trajectory.

9.1 Configuration and EOW result

The Volve case applies the generic eleven-term configuration in Table 1. The calculated envelope is therefore an illustrative output of the implemented model, not an estimate of the survey uncertainty achieved by the actual drilling operation.

Table 8: End-of-well result for Volve well 15/9-F-11 A using the default 2σ convention. Horizontal semi-axes come from the NE marginal covariance, the vertical value is $2\sqrt{C_{VV}}$.

Calculated coordinates		Modelled uncertainty	
MD	3762.00 m	NE semi-major	11.00 m
TVD	3126.49 m	NE semi-minor	3.52 m
North	706.04 m	NE angle	119.50°
East	297.20 m	Vertical extent	± 11.62 m

The NE angle is measured clockwise from North toward East in the internal NE coordinate order. The vertical value is a 2σ coordinate extent, not the vertical extent of a joint three-dimensional probability region. Probability mode labels these quantities separately and applies the dimension-specific factors stated above.

9.2 Trajectory segmentation

For descriptive analysis, the well is divided into the five geometric sections in Table 9. The boundaries were selected from changes in inclination, azimuth, and DLS and should not be interpreted as formal drilling phase classifications.

This subsection defines the geometry only. The later “Attitude and uncertainty response” subsection intentionally reuses the same five boundaries to interpret the covariance components. Keeping those two tasks separate avoids mixing observed trajectory shape with conclusions about uncertainty propagation.

Table 9: Descriptive trajectory sections used in the case-study discussion.

Section	End MD (m)	End TVD (m)	Geometric description
1. Near-vertical	620.0	619.8	Low inclination before directional change.
2. 3D turn	2130.0	2029.3	Predominantly westward displacement followed by an eastward correction, inclination returns below approximately 3°.
3. High-inclination build	3168.1	2755.8	northeastward displacement with inclination increasing to approximately 75°.
4. Inclination drop	3571.7	2980.7	Inclination decreases from approximately 75° to 40°.
5. 40-degree hold	3762.0	3126.5	Approximately constant 40° inclination to TD.

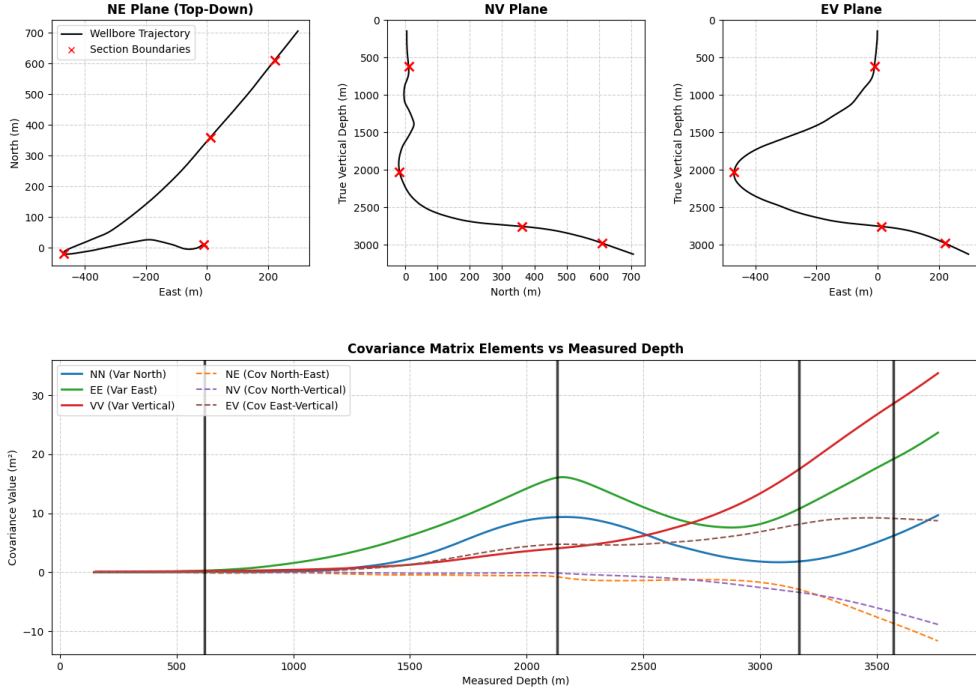


Figure 5: Trajectory projections and covariance elements using one consistent set of five descriptive section boundaries.

9.3 Attitude and uncertainty response

With the geometry established above, this subsection examines how the covariance changes within those sections. The local sensitivity directions explain the main qualitative changes:

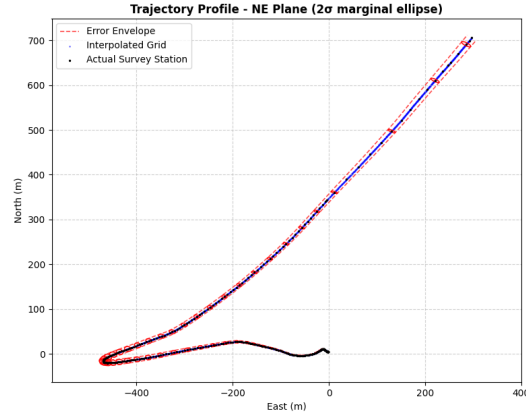
- At low inclination, MD errors primarily affect TVD, inclination errors produce lateral displacement, and azimuth sensitivity is suppressed by $\sin I$.
- At high inclination, MD and azimuth errors primarily affect the horizontal position, while inclination errors have a strong vertical component.
- Random contributions add positive-semidefinite outer products and do not cancel. Correlated error vectors can partially cancel when their NEV direction changes along the trajectory. Consequently, a diagonal element or the trace of the total covariance can decrease between stations even though every covariance matrix remains positive semidefinite.

The sectionwise observations are summarized below. They describe the generated eleven-term result and are not universal properties of wells with similar shapes.

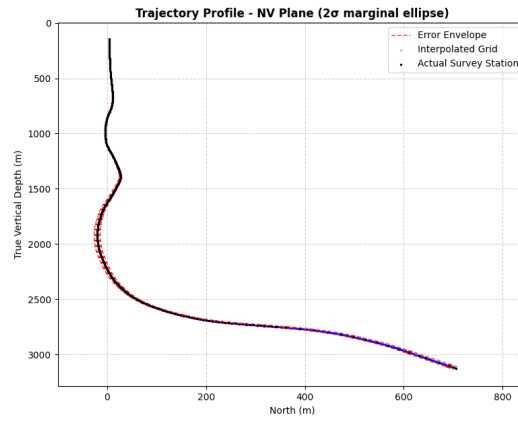
- **Section 1:** Covariance remains small because displacement is limited and the selected angular terms have little accumulated positional leverage near the start of the well.
- **Section 2:** C_{EE} grows more strongly than C_{NN} during the predominantly westward displacement. The term-level decomposition indicates that SAGE is the main contributor to the observed early C_{EV} behaviour.

- **Section 3:** Direction changes allow components of accumulated correlated error vectors to partially cancel, reducing their contributions to C_{EE} and C_{NN} , while random contributions retain a non-cancelling baseline. C_{VV} increases as the trajectory approaches high inclination.
- **Section 4:** The inclination decrease changes the vertical and lateral sensitivity projections. The observed C_{EV} trend is consistent with the changing accumulated East and vertical components of the dominant systematic terms.
- **Section 5:** Under approximately constant attitude, the generated covariance components return to a smoother growth pattern.

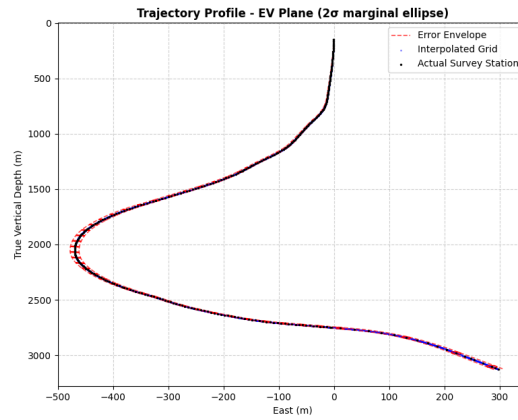
9.4 Planar and three-dimensional visualization



(a) NE map view.



(b) NV vertical projection.



(c) EV vertical projection.

Figure 6: Trajectory and 2σ marginal covariance ellipses in the three global coordinate planes. Plot aspect ratios must be considered when visually interpreting ellipse shapes.

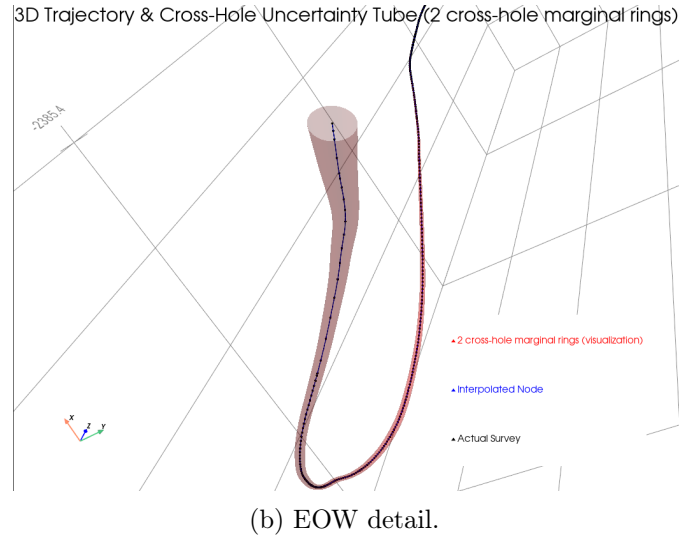
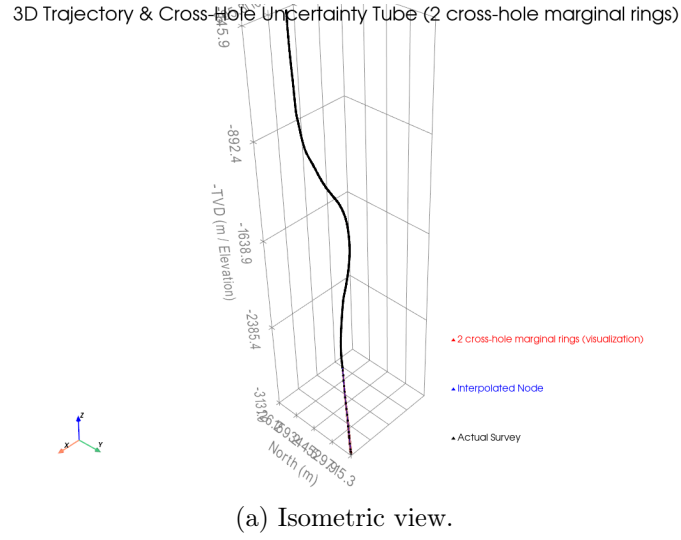


Figure 7: Three-dimensional trajectory shown from two viewpoints with the default 2σ local cross-hole uncertainty tube. The surface is built from normal-plane covariance rings and is not an operational anti-collision volume.

9.5 Variance-growth diagnostic

The diagnostic quantity

$$g_k = \frac{\text{tr}(\mathbf{C}_k) - \text{tr}(\mathbf{C}_{k-1})}{D_k - D_{k-1}} \quad (33)$$

is plotted with course length, DLS, inclination, and azimuth. It measures the station-to-station change in total coordinate variance per metre. It is not an ellipsoid-volume derivative. For a fixed scaling factor, ellipsoid volume is proportional to $\sqrt{\det \mathbf{C}}$.

Negative g_k values are possible when cancellation in correlated error vectors exceeds the positive increment from random contributions. This does not imply negative variance, only a decrease in the trace between consecutive stations.

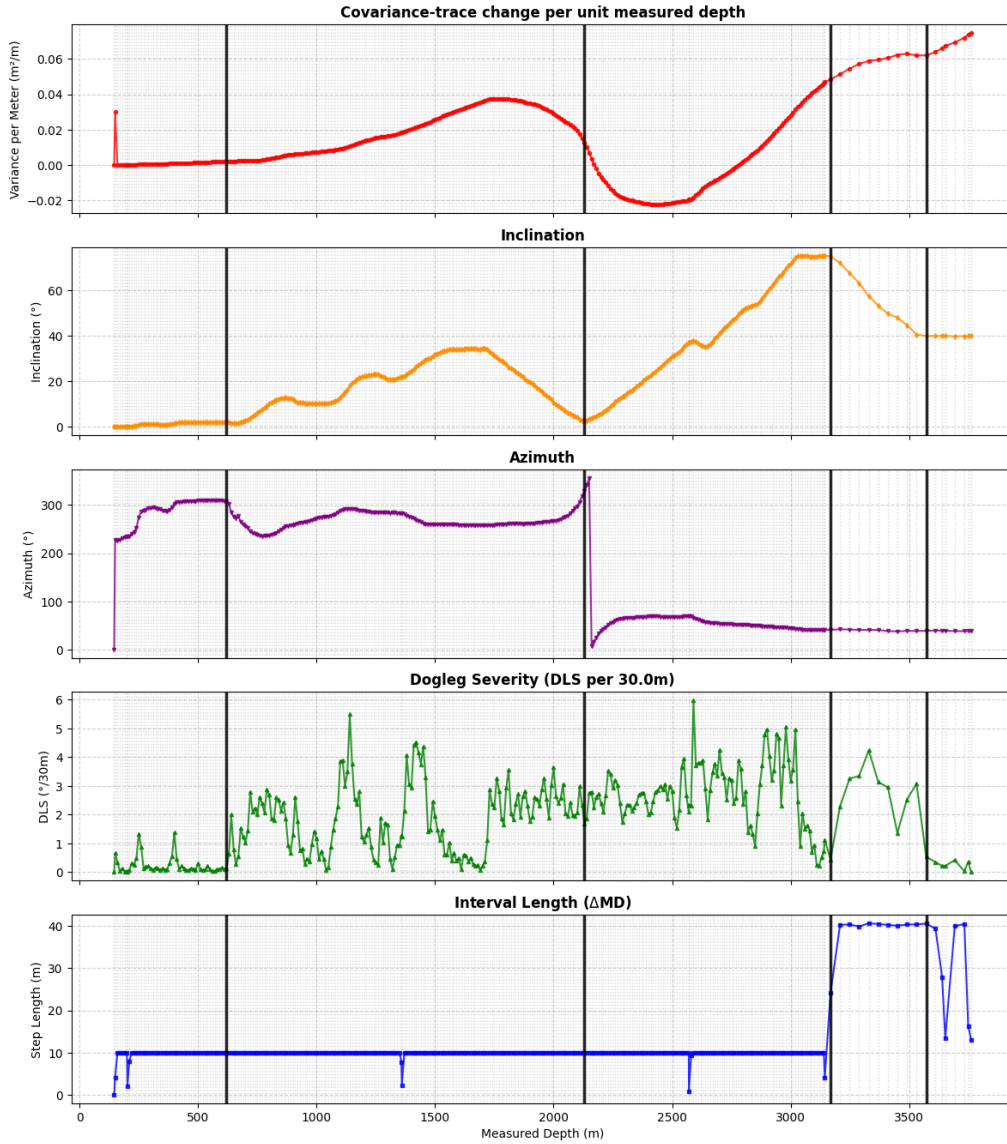


Figure 8: Change in covariance trace per metre together with measured-depth interval, DLS, inclination, and azimuth. Associations in this figure are used as diagnostics and are not treated as isolated causal tests.

10 Engineering Assessment and Limitations

10.1 Supported conclusions

The combination of benchmark and finite-difference tests supports the following claims:

1. The trajectory implementation is numerically consistent with the supplied ISCWSA and Volve coordinates at centimetre-to-millimetre scale.
2. The selected BT covariance implementation reproduces the reference spreadsheet values for ten terms approximately within numerical round-off.
3. The analytical BT and MCM sensitivity implementations reproduce their corresponding numerical derivatives for the tested intervals.
4. For the selected Volve geometry and eleven-term configuration, the sensitivity choice has a small effect on the complete covariance, projected axes, vertical marginal extent, 3D principal axes, and NE orientation.

These tests are deliberately complementary. Spreadsheet agreement verifies the selected ISCWSA calculation path, while finite differences verify the derivative implementation independently of the benchmark covariance values.

10.2 Model and validation limitations

The work does not demonstrate that the framework is a complete or certified ISCWSA implementation. It does not establish the physical positional accuracy of the Volve survey, prove MCM and BT equivalence for other geometries, or validate use of the generated envelope in collision avoidance.

The small MCM–BT difference observed here should be understood in the context of the selected error terms and survey spacing. Other PUMs, larger doglegs, different inclinations, longer survey intervals, or additional sources may produce a different comparison.

10.2.1 Model scope

- Only eleven selected terms are implemented. Magnetic- and gravity-field-dependent sensor terms are excluded.
- The Volve case uses generic benchmark magnitudes rather than a tool-specific PUM for the actual survey system.
- Surface-position and tie-on uncertainty are not included.
- Multi-leg propagation, well-by-well correlations, global correlation between wells, separation factor, and anti-collision calculations are not implemented.
- Magnetic, true, and grid north transformations are assumed to have been completed before input.

10.2.2 Statistical and numerical scope

- Covariance describes only second-order statistical structure and does not represent skewness, multimodality, or non-Gaussian tails.
- The 2σ display convention assumes that standard-deviation scaling is meaningful. Optional probability regions additionally rely on a multivariate Gaussian/chi-square interpretation.

- Finite differences depend on the perturbation scale, although the selected scales were checked by convergence sweeps.
- Very small negative eigenvalues may arise from round-off and require a documented numerical tolerance rather than unconditional clipping.
- Covariance interpolation is used only for visualization and is not a substitute for physical survey observations.

10.2.3 Validation scope

- The remaining spreadsheet TVD differences cannot be assigned to a unique cause because not all reference-calculation details are exposed.
- Volve reference coordinates are rounded and are not independent ground truth.
- Only one field trajectory is used for the propagated MCM–BT comparison.
- Although the comparison includes the complete covariance, projected axes, 3D principal axes, and NE orientation, it remains specific to one field trajectory and the selected eleven-term model.

10.3 Future work

The next technically useful extensions are:

1. controlled synthetic trajectories spanning dogleg, inclination, azimuth change, and survey spacing
2. completion of an appropriate Revision 5 PUM with field-dependent reference quantities
3. explicit multi-leg, well-by-well, and global correlation handling
4. relative covariance and separation-factor calculations for a second well

11 Conclusion

This project implemented an end-to-end numerical path from directional survey stations to trajectory coordinates, selected ISCWSA position covariance, and two- and three-dimensional uncertainty visualizations.

The Minimum Curvature trajectory calculation reproduced the available reference coordinates with maximum TVD residuals of 35.9 mm in the ISCWSA examples and a maximum component residual of 8.70 mm against the rounded Volve coordinates. These values demonstrate agreement with the supplied references, not physical survey accuracy.

Using the BT sensitivities documented by ISCWSA, ten selected covariance terms matched the reference spreadsheets at approximately floating-point precision. The remaining DSTG difference was below $9.3 \times 10^{-5} \text{ m}^2$ and is consistent with its dependence on the slightly different reconstructed TVD.

Central finite differences reproduced both the analytical BT and analytical MCM sensitivities for the tested trajectory intervals. The MCM formulation retains the ratio factor and its angular derivatives, whereas BT omits them. For Volve well 15/9-F-11 A and the selected eleven-term configuration, the maximum NE semi-major-axis difference was 1.559 mm, the maximum axial NE-orientation difference was 0.004187° , and the complete covariance had a global BT-referenced relative RMS difference of 1.244×10^{-4} . The result is small for this case, not evidence of general equivalence.

The implementation is therefore a numerically verified educational implementation of the tested eleven-term subset and a basis for further work on complete PUMs, multi-well correlation, and anti-collision calculations. Its main value is the transparent connection between trajectory geometry, analytical sensitivities, covariance propagation, numerical verification, and spatial visualization.

Code and Data Availability

Source code, execution instructions and used data-sets are available at:

<https://github.com/jczettl/wellbore-trajectory-uncertainty>

A Complete Covariance Diagnostic

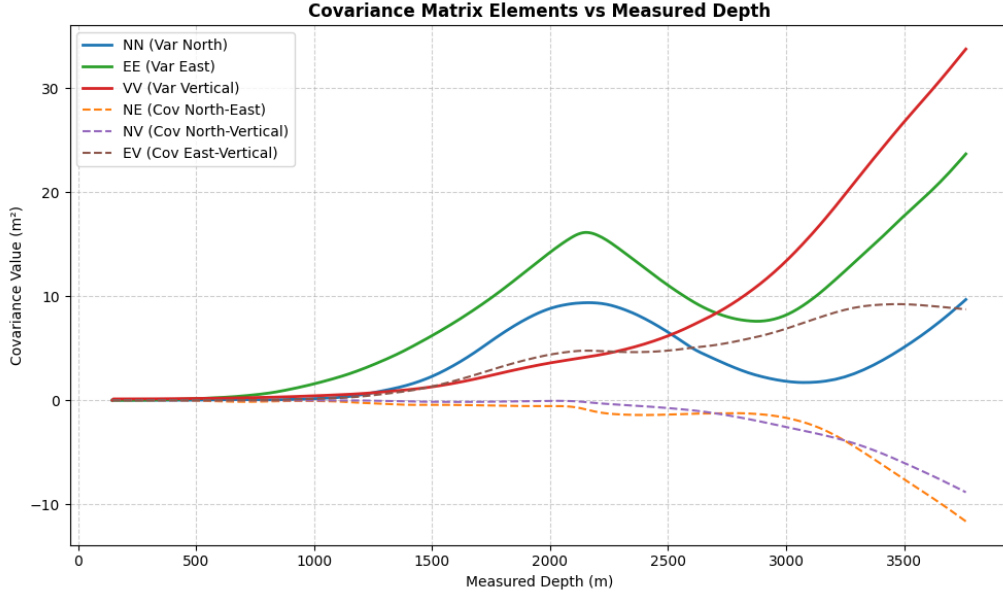


Figure 9: All unique NEV covariance elements versus measured depth for the Volve case. The off-diagonal signs describe covariance and do not by themselves imply invalid uncertainty.

B Term-Level EV Diagnostic

The total C_{EV} curve was decomposed by error source to test the proposed geometric explanation. In the selected model, SAGE is the principal contributor to the early EV behaviour. Because SAGE is systematic in this configuration, its covariance contribution at a station is the product of the accumulated East and vertical error-vector components,

$$C_{EV}^{\text{SAGE}} = E_E^{\text{SAGE}} E_V^{\text{SAGE}}. \quad (34)$$

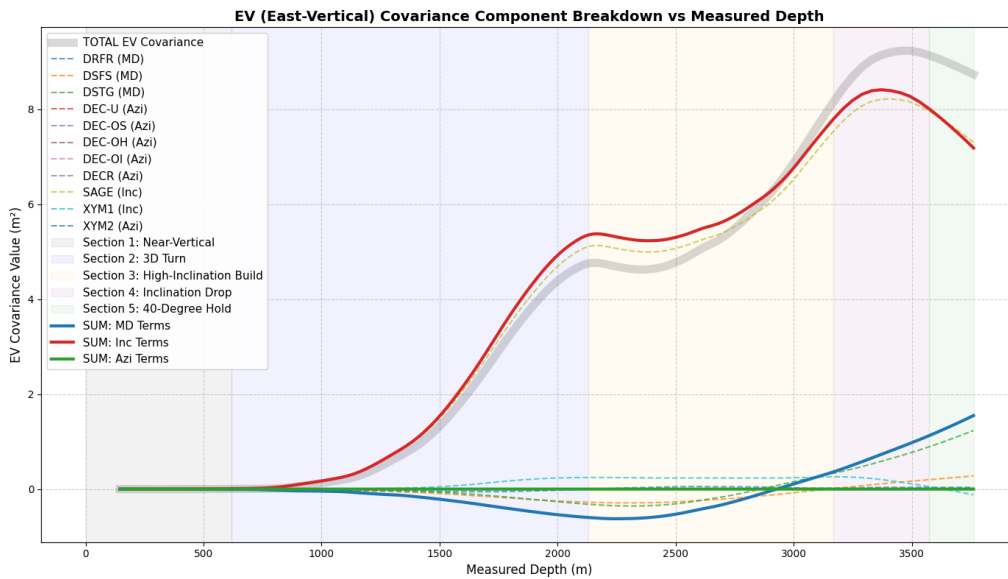


Figure 10: East–Vertical covariance contribution by error term. The corrected SAGE vector-product identity is covered by the automated anchor-station test.

C MCM-versus-BT Covariance Elements

Table 10 resolves the full-matrix comparison into the six independent NEV covariance elements. The RMS and maximum columns contain absolute MCM–BT differences in square metres. Relative RMS uses the complete BT profile of each element as one denominator, avoiding unstable pointwise percentages where an off-diagonal element crosses zero. The EOW column retains the sign of MCM minus BT.

Table 10: MCM-versus-BT comparison for the six independent covariance elements. Relative RMS uses the BT element profile as one global reference scale, EOW differences are signed MCM minus BT.

Element	Unit	RMS	Maximum	Relative RMS	EOW difference	MD at max. (m)
C_{NN}	m ²	7.222×10^{-4}	4.459×10^{-3}	1.539×10^{-4}	4.459×10^{-3}	3762.00
C_{EE}	m ²	1.008×10^{-3}	5.733×10^{-3}	1.093×10^{-4}	5.733×10^{-3}	3762.00
C_{VV}	m ²	6.483×10^{-4}	1.630×10^{-3}	7.733×10^{-5}	1.309×10^{-3}	3168.10
C_{NE}	m ²	6.468×10^{-4}	3.677×10^{-3}	3.083×10^{-4}	-3.677×10^{-3}	3762.00
C_{NV}	m ²	3.356×10^{-4}	1.867×10^{-3}	1.899×10^{-4}	-1.867×10^{-3}	3762.00
C_{EV}	m ²	4.917×10^{-4}	1.096×10^{-3}	1.193×10^{-4}	-1.096×10^{-3}	3762.00

The largest absolute element difference is 5.733×10^{-3} m² in C_{EE} at EOW. The largest relative RMS is 3.083×10^{-4} for C_{NE} , so every element remains below 0.031% on its global BT reference scale. Positive EOW differences in the three variances mean that the MCM result is slightly larger for those components, whereas the negative off-diagonal differences represent small changes in covariance orientation and coupling. Their signs do not indicate better or worse positional accuracy.

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- [1] Industry Steering Committee on Wellbore Survey Accuracy, *Definition of the ISCWSA Error Model*, Revision 5.13, January 2023. Available at: <https://www.iscwsa.net/files/842/>.
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- [3] H. S. Williamson, “Accuracy Prediction for Directional Measurement While Drilling,” *SPE Drilling & Completion*, vol. 15, no. 4, pp. 221–233, 2000, SPE-67616-PA.
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